

WENRUI HUANG

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EDUCATION	Brown University <i>Ph.D. in Mathematics</i> • Advisor: Benoît Pausader	Providence, RI 2022 - 2027 (<i>expected</i>)
	University of Science and Technology of China (USTC) <i>B.S. in Mathematics</i> • Advisor: Yong Wei • Thesis: Mean curvature flow with surgery and its applications	Hefei, China 2018 - 2022
RESEARCH INTERESTS	Partial differential equations, nonlinear dispersive PDEs and kinetic equations, stability and long-time behavior.	
PAPERS AND PREPRINTS	1. <i>Long-Time Dynamics and Modified Scattering for the Coulomb Vlasov-Hartree System</i> , with M. Xie. arXiv:2608.04402. 2. <i>Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential</i> , with H. Kwon. arXiv:2602.21344. 3. <i>Stability and instability of small BGK waves</i> , with D. Bian, E. Grenier, and B. Pausader. arXiv:2601.10030. 4. <i>The Vlasov-Poisson system with a perfectly conducting wall: Convex domains</i> , with B. Pausader and M. Suzuki. To appear in Comm. Math. Phys. 5. <i>Modified wave operators for the Hartree equation with repulsive Coulomb potential</i> , arXiv:2411.02737. 6. <i>Scattering of the Vlasov-Riesz system in three dimensions</i> , with H. Kwon. Annales Henri Poincaré (2026). arXiv:2407.16919.	
TALKS	• GLESPA Seminar, Brown University • Poster Session, Workshop on Kinetic Theory and Fluids, University of Wisconsin–Madison • Brown–Yale PDE Seminar • GLESPA Seminar, Brown University • Brown–Yale PDE Seminar	Apr. 2026 Mar. 2025 Apr. 2024 Apr. 2024 Nov. 2023
TEACHING	Brown University • TA, MATH 0180 Calculus III, Fall 2026 • TA, MATH 0190 Calculus II (Physics/Engineering), Fall 2025 • TA, MATH 0180 Calculus III, Spring 2024 • TA, MATH 0180 Calculus III, Fall 2023 • TA, MATH 0100 Calculus II, Spring 2023 University of Science and Technology of China • TA, Riemannian Geometry, Spring 2022 • TA, Differential Geometry, Fall 2021 • TA, Topology, Spring 2021 and Spring 2022	